

KOHLER AI BathPlan

The AI can imagine a bathroom. The constraint engine decides whether it survives reality.

The problem

People planning a bathroom do not have an imagination problem. They cannot answer the question that actually matters:

“Will these products fit in my room, together, within my budget — and if not, what should I give up?”

That is a configuration decision under hard constraints, not an image-generation problem. A 72-inch vanity does not fit a 5-foot wall no matter how good the render looks.

The solution

A deterministic planner wrapped in a conversation:

1. Fixtures placed against walls with the clear floor plumbing codes require
2. Budget, compatibility, power and installation checked — unknowns stay unknown
3. 2–3 options that differ in ways you would actually choose between
4. Water impact estimated with every formula and assumption shown
5. **When the requirements cannot all hold, it says so — and offers the trade-offs it actually evaluated**

33

curated products

~1,460

configurations evaluated per plan

130

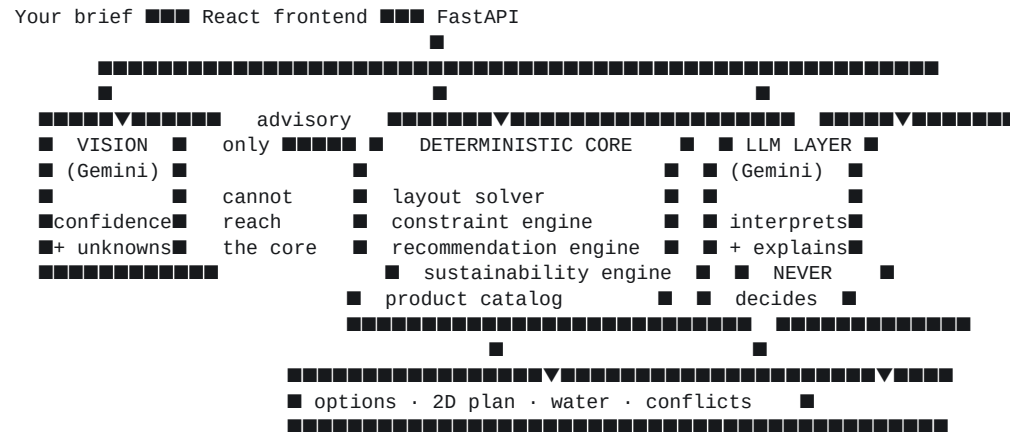
automated tests

0

figures produced by an LLM

The LLM is not the source of truth

Feasibility, dimensions, pricing and water figures are computed before any model is called. The model receives them as given facts.



Enforced structurally, not by instruction

Vision output is typed authoritative: `Literal[False]` and has no field for a dimension or rough-in — it *cannot* reach the engine.

Intent output is sanitised against a fixed vocabulary, so an invented category is dropped before the pipeline.

Every response carries `computed_by`. “**llm**” **never** appears in it.

Works with no API key

The AI layers are enhancements with tested fallbacks, not load-bearing parts. With no key configured, planning, the 2D plan, water estimation, conflict resolution and plain-language modification all still work — the full test suite runs in this mode.

A model failure degrades the wording, never the verdict.

The most important screen is the one that says no

Constraint conflict resolution

User: “Add a smart shower and a smart toilet and a bathtub, keep my budget.”

System: over budget by ■18,300 — *and* physically unplaceable in 6 × 8 ft.

It names both, and notes that **more budget would not fix a spatial problem**. It then offers the relaxations it actually evaluated and lets the user choose. The plan is recomputed from scratch, not patched.

A generative tool would have returned something plausible and let the customer discover the problem during installation.

Against a generative approach

	Typical AI design tool	BathPlan
Spatial reasoning	Implied by a rendering	Solved geometry, IRC clearances
Feasibility	Asserted	Computed; failing check named
Unknowns	Filled in plausibly	Stay unknown, surfaced
Impossible ask	Something close, presented as the answer	Refused, explained, alternatives
Water savings	A percentage	Formula + baseline + assumptions
LLM role	Decides	Interprets and explains

Confidence-aware by construction. A constraint report has three states, not two: *feasible*, *feasible pending verification*, and *infeasible*. An unmeasured rough-in is not the same as a mismatched one — conflating them either blocks every real plan or fakes certainty. Pending options are offered, ranked strictly below verified ones, and carry their outstanding checks with them.

A decision layer upstream of the design consultation

Why this matters to KOHLER

KOHLER already offers a human bathroom design service. BathPlan does not compete with it — it sits in front of it, turning an undecided browser into a customer arriving with a feasible, costed, explained configuration.

- Specification errors surface before purchase rather than at installation.
- Smart and water-efficient products are recommended on evidence the customer can audit, not on marketing adjectives.
- Every plan produces a verification list — a natural handoff to a KOHLER consultant or installer.
- The catalog is designed to be swapped for an authorised live feed with no engine change.

What we would build next

1. Authorised catalog integration — replace illustrative data with a live feed; the schema already separates verified from illustrative fields.
2. Regional code packs — swap IRC clearances for Indian standards. The rules are already isolated in one module.
3. Guided rough-in capture — convert the commonest verification requirement into a verified fact.
4. Designer handoff package — configuration, plan and open verifications, exported to a consultant.
5. Konnect-informed water tracking — replace assumed usage with measured usage.

Stated plainly: prices in this prototype are illustrative and are not KOHLER pricing. Clearances are US residential-code figures used as planning minimums, and usage benchmarks are US-derived. A photograph establishes nothing measurable. This is a planning aid, not an engineering drawing — final dimensions, plumbing, electrical and local-code requirements must be verified by a qualified professional.